Talexa Lecture Series

Generated by Talexa

March 10, 2026

Supervised Learning

- Supervised learning is a machine learning algorithm where the training dataset consists of inputs paired with their correct outputs known as ground truths.
- We aim to optimize the model's parameters by minimizing a previously defined cost function utilizing the labeled training dataset.
- The desired function f can be approximated by finding the optimal set of parameters θ^* that optimize the loss function.
- Limitations:
  - Needs experience
  - Costly and time-consuming
  - Inter or intra-observer variability

Semi-Supervised Learning

- The semi-supervised learning (SSL) framework provides the solution by utilizing a huge amount of unlabeled data along with a few annotated ones.
- To leverage both labeled and unlabeled data in the SSL paradigm, the objective function takes the form:

$$\text{Loss} = L_{\text{Supervised}} + \beta L_{\text{unsupervised}}$$

- Where $L_{\text{Supervised}}$ denotes the supervised loss, $L_{\text{unsupervised}}$ denotes the unsupervised loss, and β is a weighing factor that controls the contribution of the unsupervised loss.

Intuition of Semi-Supervised Learning

- Consider this dataset, which represents a binary classification problem.
- If we depend only on the labeled data, supervised learning updates model weights to minimize the average difference between predictions and labels. However, a valid decision boundary for the data points might not be found with limited labeled data.

Intuition of Semi-Supervised Learning

- On the other hand, unsupervised learning tries to cluster points together based on similarities in some feature space.
- But, without labels to guide training, an unsupervised algorithm might find sub-optimal clusters.

Intuition of Semi-Supervised Learning

- In the semi-supervised setting, we use labeled and unlabeled data. Our labeled points act as a sanity check; they ground our model predictions and add structure to the learning problem by establishing the number of classes and which clusters correspond to which class.
- Unlabeled data points provide context; by exposing our model to as much data as possible, we can accurately estimate the shape of the whole distribution.
- With both parts—labeled and unlabeled data—we can train more accurate and resilient models.

Techniques Used in Semi-Supervised Learning

- Self training methods: The most known algorithm is Pseudo-labeling.
- Train a model with labeled data.
- Use the trained model to predict labels for the unlabeled data, which creates pseudo-labeled data.
- Retrain the model with the pseudo-labeled and labeled data together.
- Accept only the predictions (pseudo-labels) that have a high confidence, e.g., (confidence ¿= 0.9).

Techniques Used in Semi-Supervised Learning

- Label Propagation: This technique is a graph-based transductive method for inferring pseudo-labels for unlabeled data. The labeled and unlabeled data points can be considered as nodes of a graph, and the objective is to propagate the labels from the labeled nodes to the unlabeled data points within the same dataset; in this case, it does not generalize to unobserved data by utilizing the similarity of two nodes, which is reflected by how strong the edge between the two nodes is.
- Inductive learning aims to generalize patterns from labeled data to predict unseen instances, i.e., a classifier capable of generalizing to unobserved data at test time.

Techniques Used in Semi-Supervised Learning

- Generative models: Since generative models can learn the distribution of real data from unlabeled samples, it can be used to facilitate SSL.
- Semi-Supervised GAN: is an extension of the GAN for addressing semi-supervised learning problems. The discriminator model is updated to predict K+1 classes, where K is the number of classes in the prediction problem. An additional class label is added for a new "fake" class. This involves directly training the discriminator model for both the unsupervised GAN task and the supervised classification task simultaneously.
- In a semi-supervised GAN, the discriminator learns to classify real images into their K real classes and detect fake images as class K+1, using labeled data for class learning, unlabeled data as real, and fake data to learn the extra "fake" class.

Semi-Supervised GAN

- Train D with labeled data
- When a labeled sample (x, y) arrives: True label $y \in \{1, 2, \ldots, K\}$, and train D using supervised cross-entropy.
- Train D with unlabeled real data
- Unlabeled real images should be classified as one of the K real classes, not "fake". Unlabeled real images should be any class 1...K (not fake).
- So we push the discriminator to output any real class (the first K classes): Meaning "this real sample should not be classified as fake".

Techniques Used in Semi-Supervised Learning

- Train D with fake images
- Generated images should be classified as the fake class (K+1)
- Train generator G
- The generator tries to fool D into thinking fake samples belong to any real class.

Techniques Used in Semi-Supervised Learning

- Consistency Regularization: This paradigm uses the idea that model predictions on an unlabeled image should remain the same even after adding noise/augmentation/variations. We could use Image Augmentation and Gaussian noise. The most typical architecture for these methods is the Teacher-Student model. The student is trained as usual, while the teacher's goal is to generate target predictions for the student. Examples of consistency regularization methods include π-model, Temporal Ensembling, and Mean Teacher.
- Exponential moving average (EMA) in Mean Teacher

Mean-Teacher

- In Mean Teacher, the teacher's weights at the beginning of training are a copy of the student's initial weights.
- At the end of training, we use the teacher model for evaluation and inference.
- Why Teacher? Its weights are a smoothed EMA of the student's weights. This means the teacher averages over many past student updates.
- Stability: Student weights can fluctuate due to stochastic training (SGD, batch noise). The EMA smooths out these fluctuations.
- This makes it more stable and better generalizing than the final student weights, which can be noisier.

Techniques Used in Semi-Supervised Learning

- 5. Hybrid Methods: These methods combine ideas from previous work, such as self-training consistency and regularization, with additional components for performance improvement.
- FixMatch uses cross entropy loss for both data (losses) Why?

Learning Paradigms

- Supervised Learning.
- Unsupervised Learning.
- Zero shot learning.
- Semi-supervised learning.
- One shot, N shot , Few shot learning.
- Weakly supervised learning.
- Federated Learning.
- Self-supervised learning.
- Transfer Learning. (check deep learning lecture).

What is federated learning?

- Federated learning is a decentralized approach to training machine learning models from data distributed on different devices/centers/locations. It doesn't require an exchange of data from client devices to global servers. Instead, the raw data on edge devices trains the model locally. The final model is formed in a shared manner by aggregating the local updates.
- Why federated learning is important?
  - Privacy: FL allows for training to occur locally on the edge device, preventing potential data breaches.
  - Data security: We can use the differential privacy or homomorphic encryption to encrypt model updates are shared with the central server, assuring data security.
  - Access to several data: FL guarantees access to data spread across multiple devices, locations, and organizations. It makes it possible to train models on sensitive data, such as financial or healthcare data while maintaining security and privacy. And thanks to greater data diversity, models can be made more generalizable.

How does federated learning work?

At every round:

- A generic baseline model is stored at the server. The copies of this model are shared/broadcasted with the client devices.
- Then train the local models based on the local data.
- Next, the updates (model parameters) from the locally trained models are shared/sent to the main model located at the server using secure aggregation techniques.
- The server/main model combines and aggregate (averages) different inputs to generate a new model using an algorithm such FedAvg.
- Once the central model has been updated on new parameters, it's shared with the client devices again for the next iteration (round).

The above five steps represent one round.

How does federated learning work?

- The models are trained for many rounds.
- With every round, the models gather a varied amount of information and improve further without creating privacy breaches.
- Over time, the models on individual devices become personalized and provide a better user experience. Since the data is collected from diverse sources, there is greater scope for the model to become generalizable.
- Round vs Epoch vs Iteration?

How does federated learning work? (FedAvg)

- Usually, at every round, a random sample of clients are selected for training. You can choose all clients.
- We perform a weighted average of the clients' updates depending on the client size.
- Represent learning algorithms such as stochastic gradient descent or any variation (Adam).
- There are many aggregation algorithms other than FedAvg such as FedSGD, FedProx, FedBN,...

Applications of federated learning

- Smartphones: Word prediction, face recognition for logging, or voice recognition while using Siri or Google Assistant are all examples of federated-learning-based solutions. It helps personalize the user experience while maintaining privacy.
- Transportation: Self-driving cars use computer vision and machine learning to analyze the surroundings and interpret the learning in real time. To continuously adapt to the environment, models need to learn from diverse datasets to improve precision.
- Manufacturing: Another example is using federated learning in industrial environment monitoring. Federated learning makes it easier to perform an analysis of the industrial environment factors obtained using multiple sensors and companies while maintaining confidential data's privacy.

Applications of federated learning

- Healthcare: The sensitive nature of healthcare data and its restricted access due to privacy issues make it difficult to scale machine learning systems in this industry globally.
- With federated learning, models can be trained through secure access to data from patients and medical institutions while the data remains at its original premises. It can help individual institutions collaborate with others and makes it possible for the models to learn from more datasets securely.
- Additionally, federated learning can allow clinicians to gain insights about patients or diseases from wider demographic areas beyond local institutions and grant smaller rural hospitals access to advanced AI technologies.

Challenges of federated learning

- Communication efficiency: represents a bottleneck of FL.
- Privacy and data protection: some of the biggest concerns with federated learning. Although the local data stays on the user's device, there's a risk for the information to be revealed from the model updates shared in the network. Common privacy-preserving techniques include:
  - Differential privacy—adding noisy data that makes it difficult to discern real information in case of data leaks
  - Homomorphic encryption—performing computation on encrypted data
- Systems heterogeneity: With the large number of devices playing a role in federated learning networks, accounting for differences in storage, communication, and computational capabilities is a huge challenge. Additionally, only a few of these devices participate at a given time, which may lead to biased training.
- Statistical (data) heterogeneity: This problem is posed by the multiple variations of data present across the client devices. For example, some devices may have high-resolution image

i.i.d= independent and identically distributed

- Independent: Each data point is independent of all other data points. This means that the occurrence or value of one data point does not affect the occurrence or value of another data point.
- Identically Distributed: Each data point is drawn from the same probability distribution. This implies that all data points come from the same population and follow the same underlying distribution, with the same parameters. If a dataset is IID, each data point is collected randomly and has the same statistical properties as all other data points. This assumption is fundamental in many statistical and machine learning models, allowing for reliable generalization and inference based on the data. However, it's important to note that data may not always adhere to the IID assumption in real-world scenarios.

Self-Supervised Learning

- The concept of self-supervised learning traces its roots to the broader domains of unsupervised learning and representation learning.
- While unsupervised learning aims to model the underlying structure of data without explicit supervision, representation learning focuses on learning effective representations of the input data.
- Self-supervised learning merges these principles, aiming to extract informative features from unlabeled data.
- In SSL, tasks fall into two categories: pretext tasks and downstream tasks.
- The pretext task: trains the model to learn meaningful representations of data. Those learned representations can subsequently be used as input to a downstream task.
- downstream task is a supervised task; classification or segmentation, or object detections. Reusing a pre-trained model on a new task is called "transfer learning."

Self-Supervised Learning Techniques

- The pretext tasks can be predictive, generative, and contrastive.
- Predictive tasks such as relative position prediction, Jigsaw puzzle, rotation prediction, and predict the applied transformation to the image.
- Generative tasks include image denoising, image inpainting, image colorization, and image reconstruction.
- Contrastive-based tasks are to contrast similar (positive) and dissimilar (negative) pairs. Positive examples are generated by applying a set of random augmentations to an input image, resulting in two transformed views of the same image, while negative examples are any other images different from the altered views.

Self-Supervised Learning Techniques

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Self-Supervised Learning Techniques

- A contrastive learning approach trains a model to distinguish between similar/positives and dissimilar/negative pairs of data points. The goal is to learn a representations where similar data points are mapped close together and dissimilar points are far apart.
- In below formula, a positive sample (x^+) is a data input with the same distribution as the image. In contrast, a negative sample (x^-) is in a different distribution to the image.

Self-Supervised Learning Techniques

- Contrastive Learning: Positive pairs could be two different views of the same image or two different augmentations of the same image. While negative pairs could be two different images.
- Contrastive Loss: Contrastive loss functions minimize the latent embedding distance between positive pairs while maximizing the distance between negative pairs. Different functions have been utilized for contrastive learning, including InfoNCE loss. InfoNCE is commonly used for contrastive loss because it corresponds to cross-entropy loss, which estimates the information shared by a pair of images. It discriminates a positive pair (zi, z+) from its related k negative pairs, written as (zi, z1-), (zi, z2-), . . . , (zi, zk-).
- τ = temperature parameter that controls the sharpness of the softmax. sim() is usually dot product or cosine similarity.

Self-Supervised Learning Techniques

- Contrastive Learning: To maximize mutual information between two related views (e.g., augmented versions of the same image or different modalities of the same data), while minimizing similarity with unrelated data.
- Encourages the model to pull together similar (positive) representations.
- Encourages the model to push apart dissimilar (negative) representations.
- The softmax over similarities makes this a classification problem where the positive is the correct class.